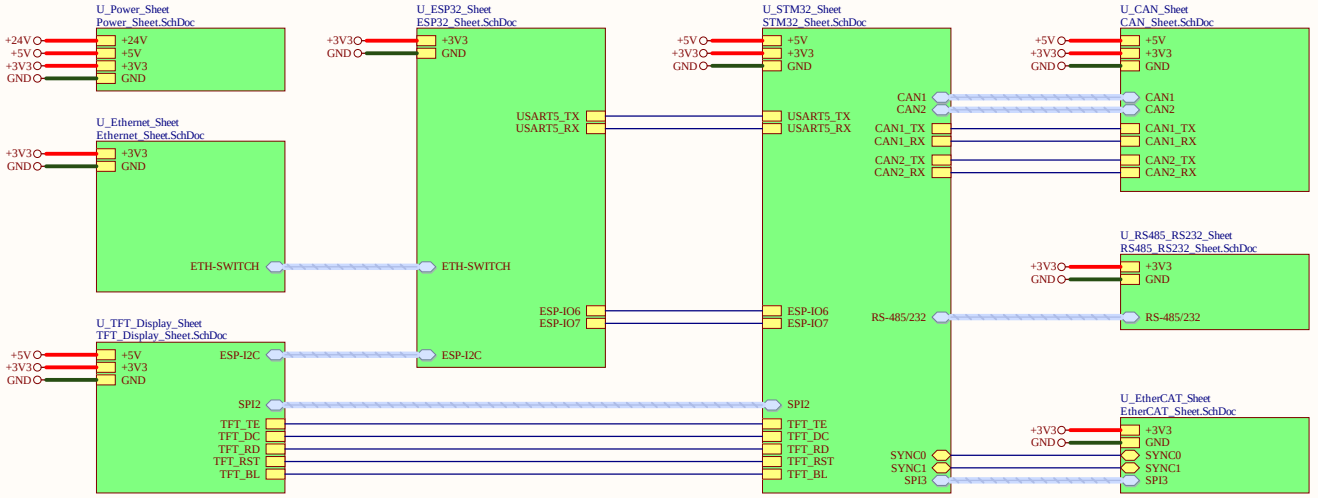
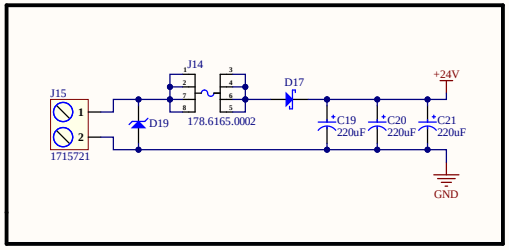
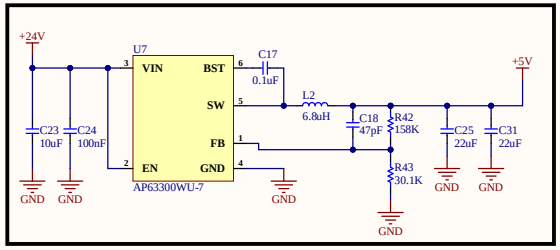


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24V INPUT

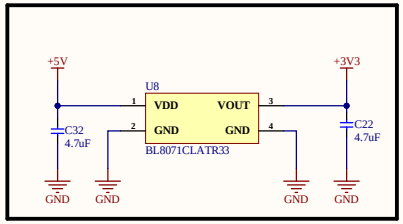


5V 3A BUCK



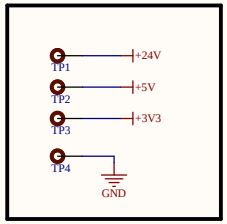
3A

3.3V 1.5A BUCK

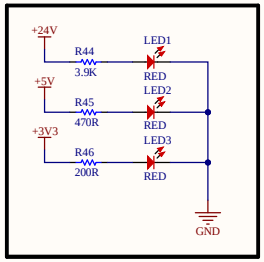


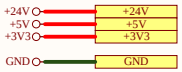
1.5A

TEST POINTS



LEDs

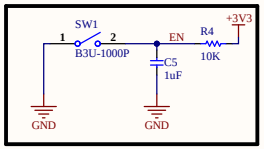


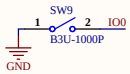


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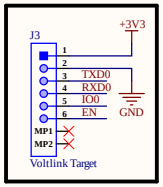
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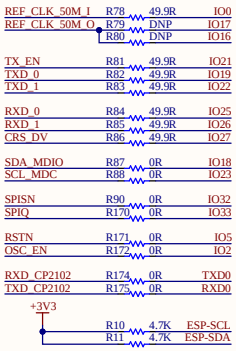
RESET

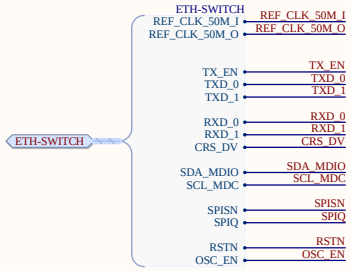


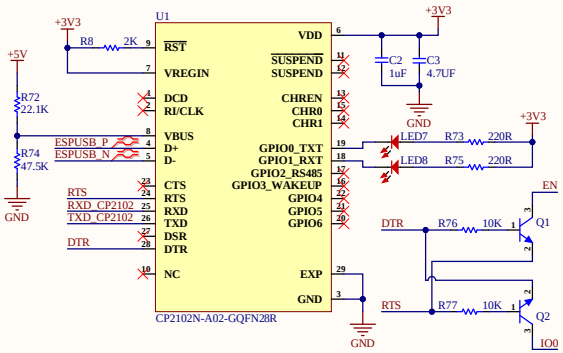


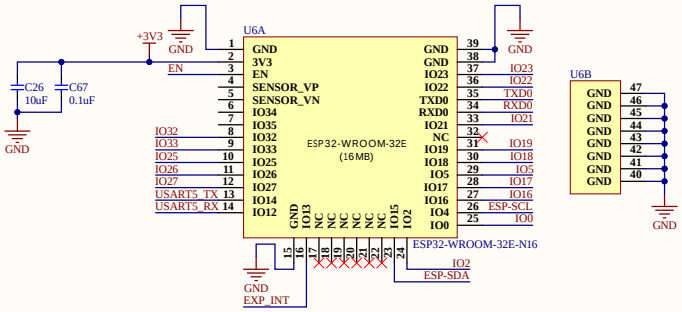
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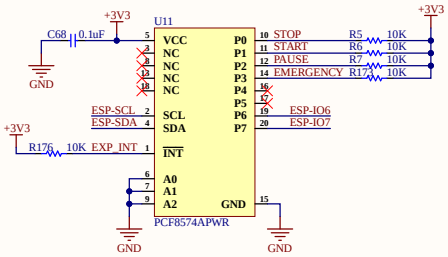


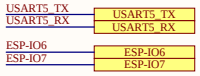






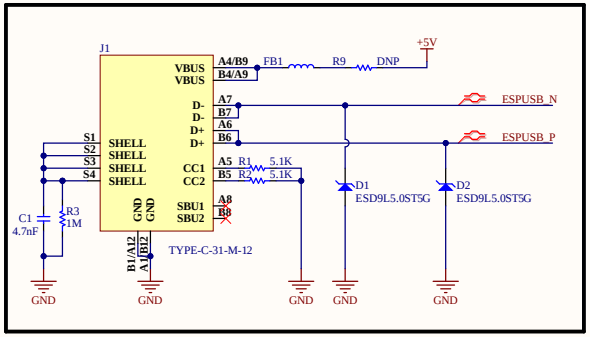




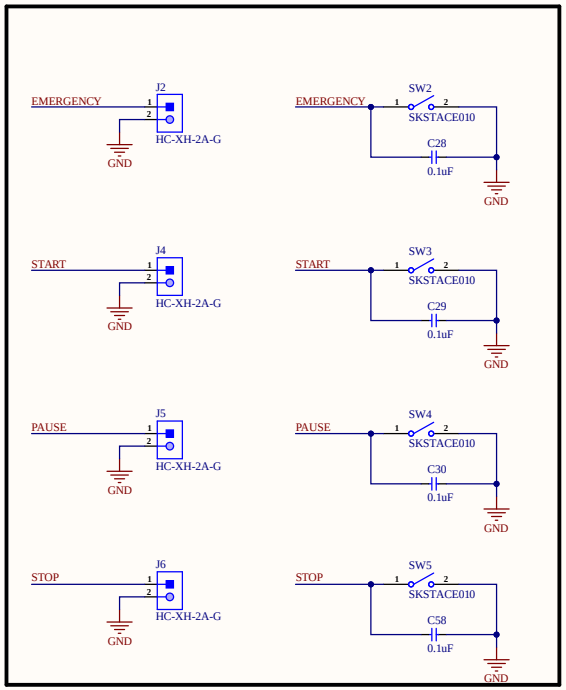




USB



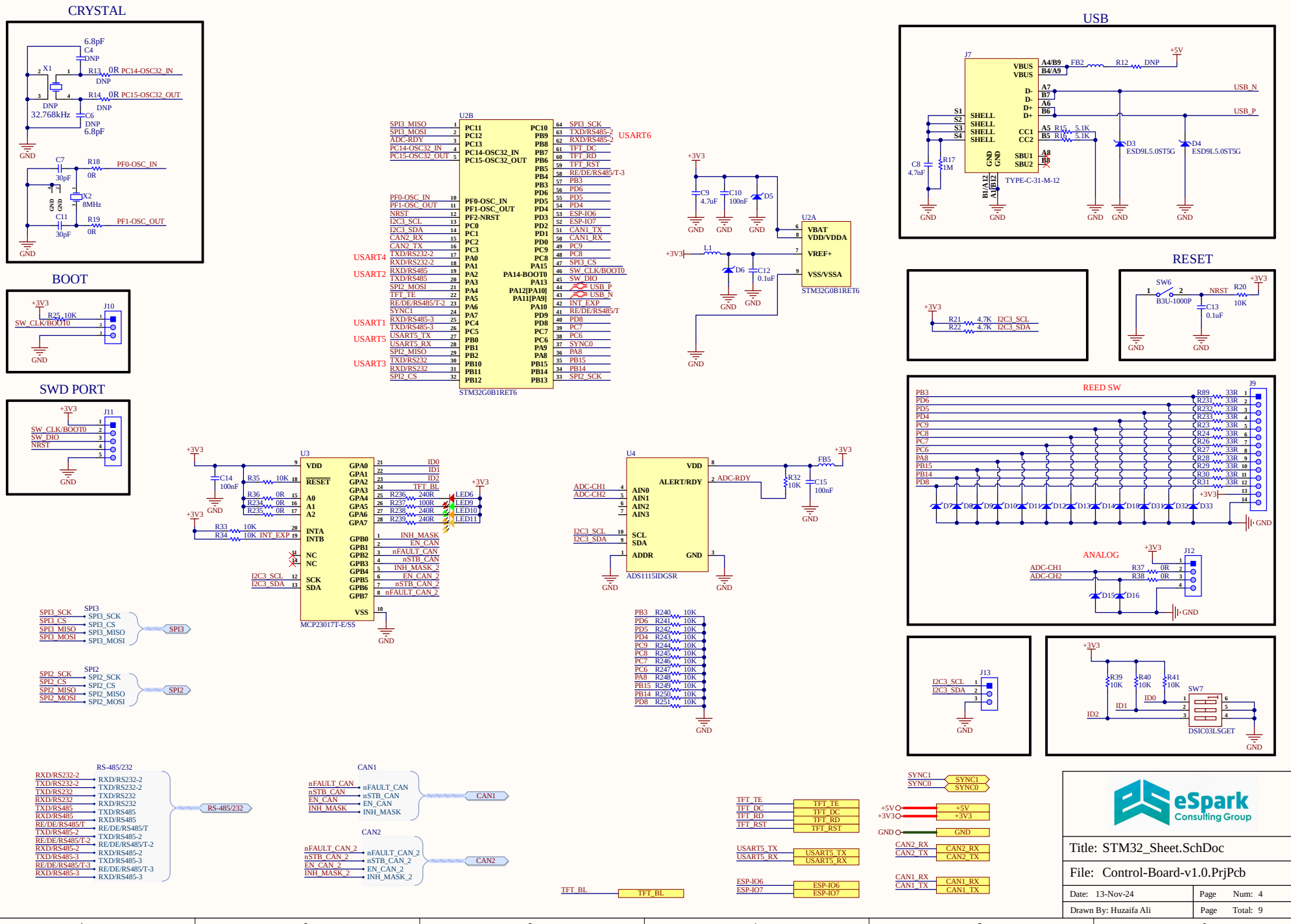
SWITCHES



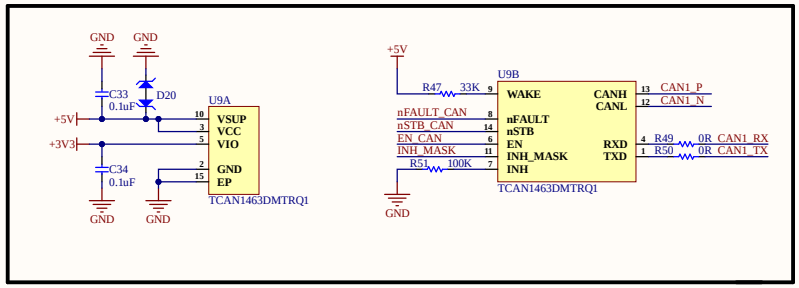


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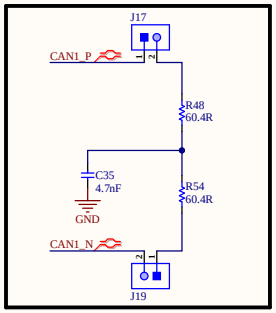
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File: Control-Board-v1.0.PrjPcb	
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CRYSTAL
BOOT
SWD PORT
USB
RESET
REED SW
ANALOG
U2B
STM32G0B1RET6
U2A
STM32G0B1RET6
U3
MCP23017T-E/SS
U4
ADS1115IDGSR
J7
TYPE-C-31-M-12
J9
J10
J11
J12
J13
SW6
SW7
RS-485/232
CAN1
CAN2
SPI3
SPI2
USART6
USART4
USART2
USART1
USART5
USART3
eSpark Consulting Group
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File: Control-Board-v1.0.PrjPcb
Date: 13-Nov-24
Page Num: 4
Drawn By: Huzaifa Ali
Page Total: 9

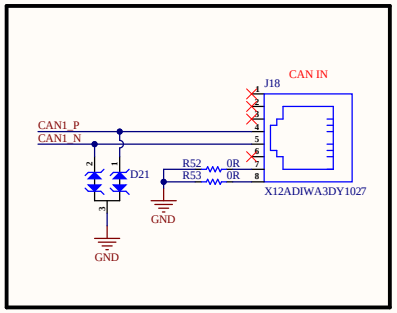
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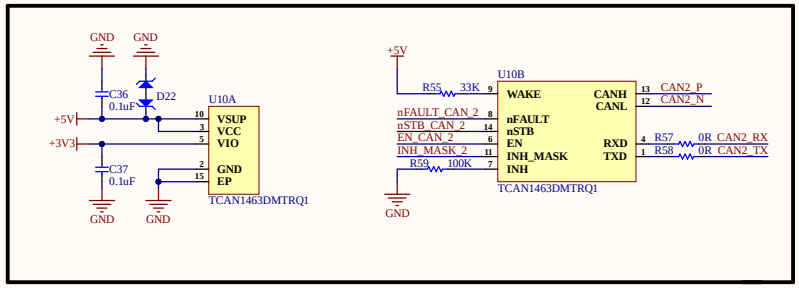
CAN TERMINATION



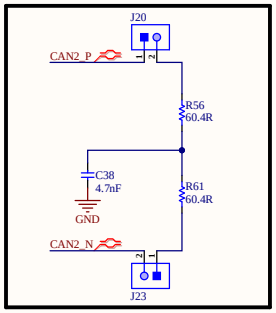
CAN CONNECTOR



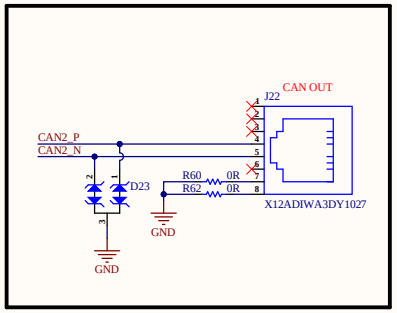
CAN BUS

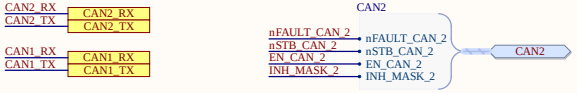


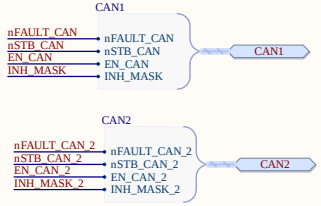
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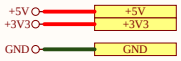


CAN2 CONNECTOR





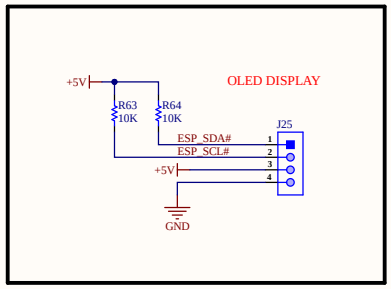




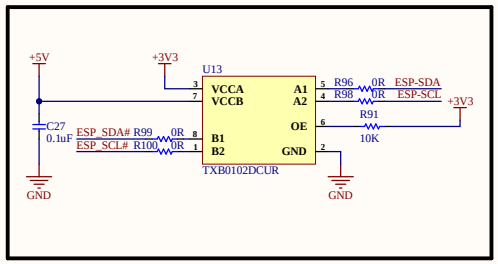
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File: Control-Board-v1.0.PrjPcb	
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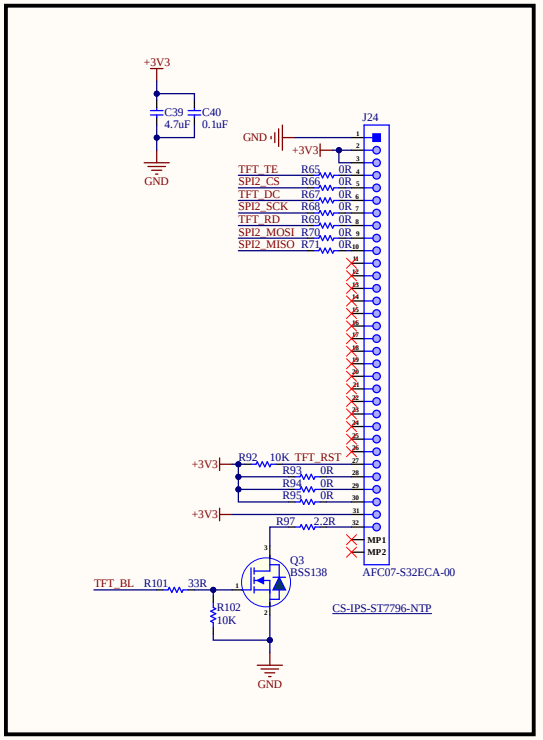
OLED



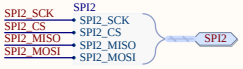
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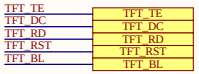


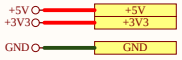
TFT LCD







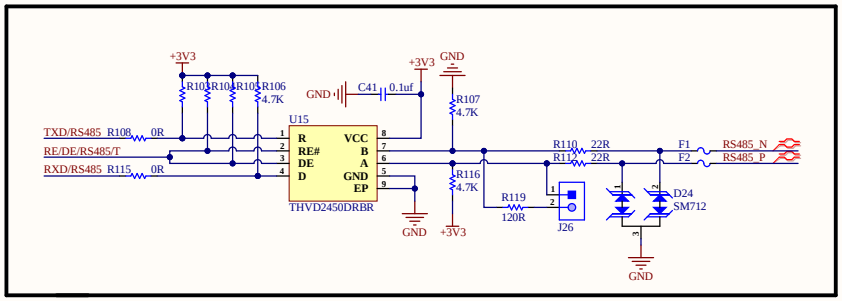




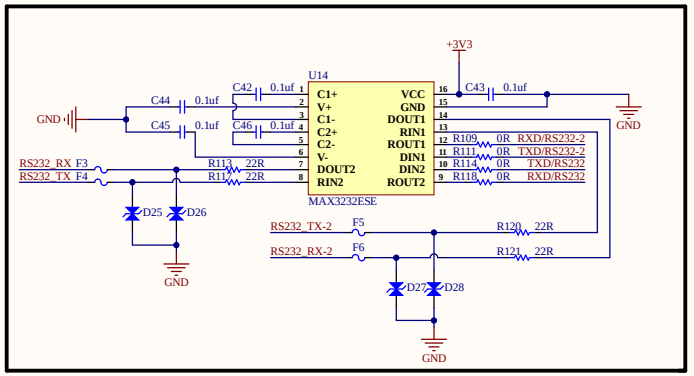
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File: Control-Board-v1.0.PrjPcb	
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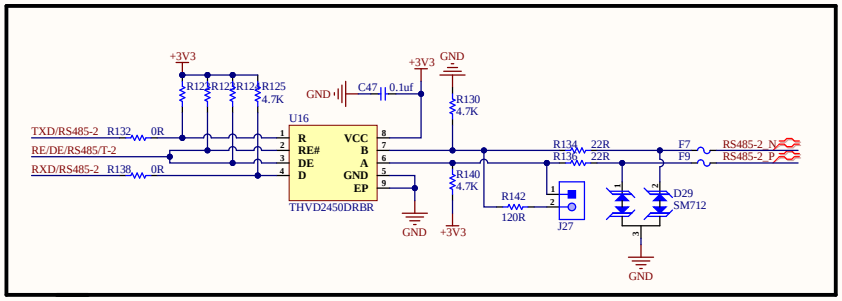
RS485 TO UART



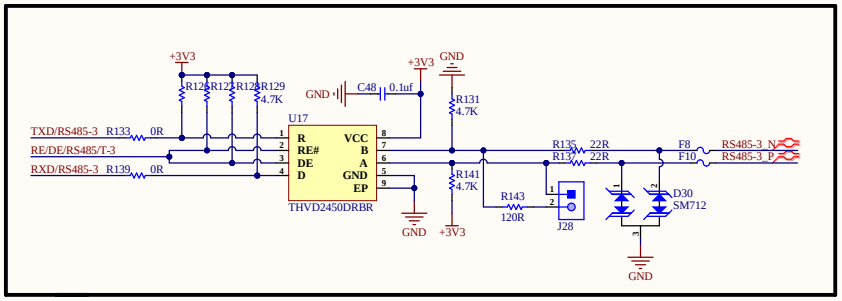
RS232 TO UART

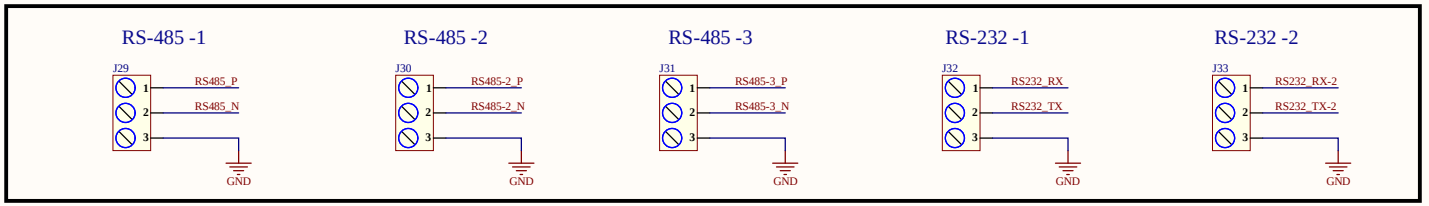


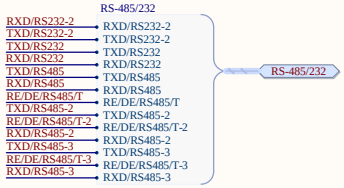
RS485 TO UART



RS485 TO UART



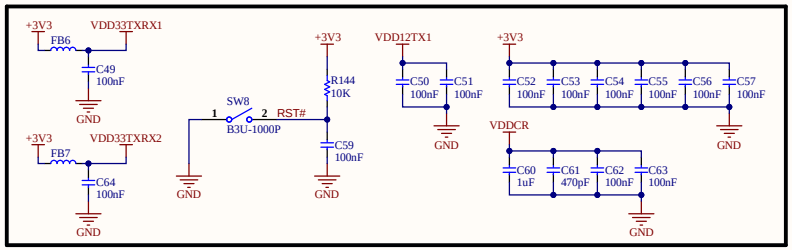




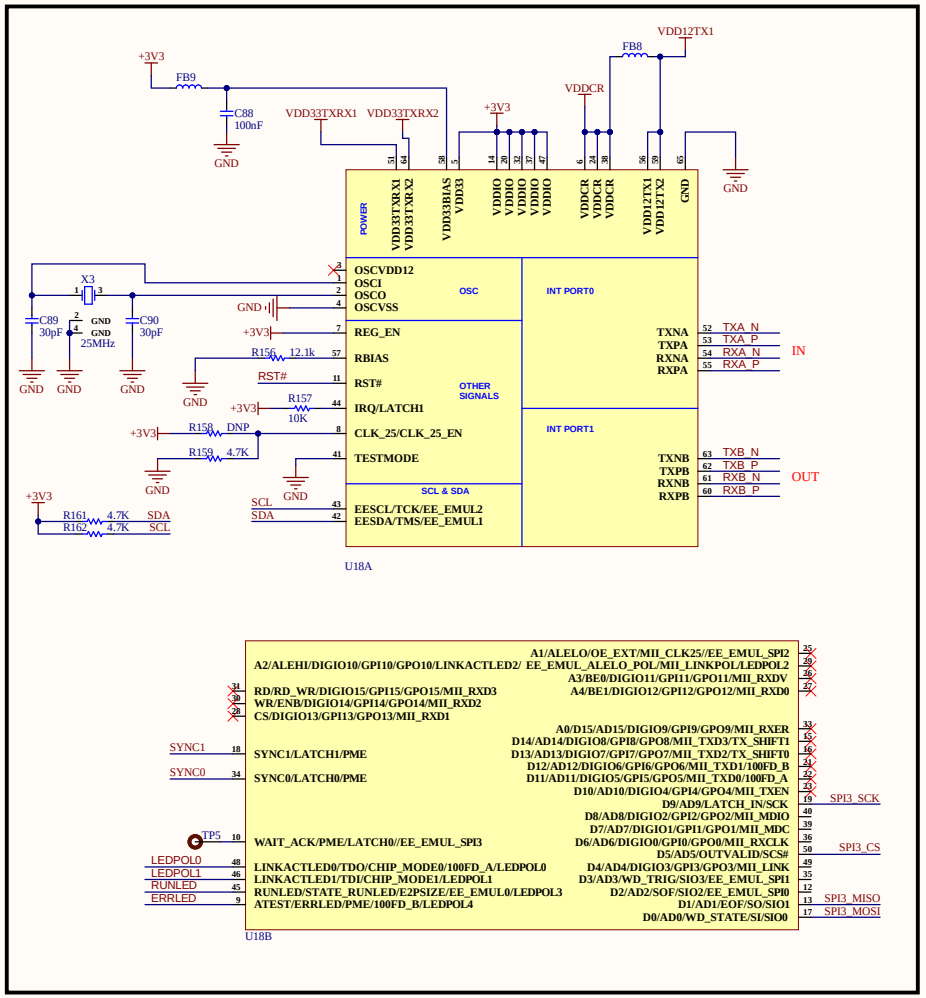


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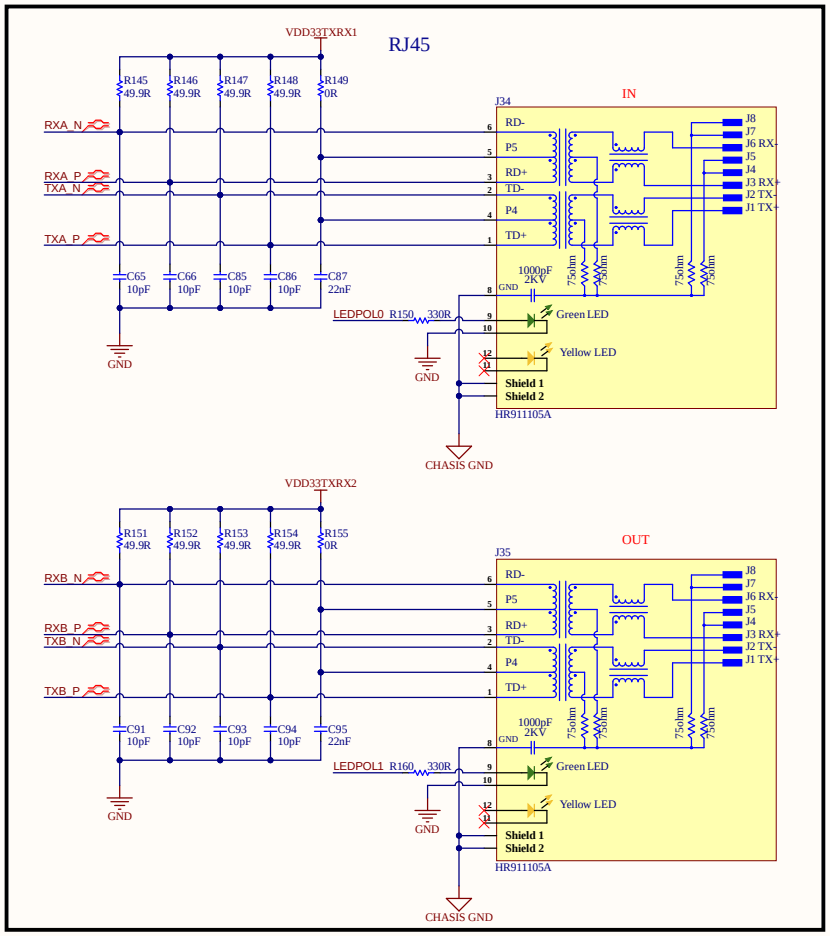


ETHERCAT

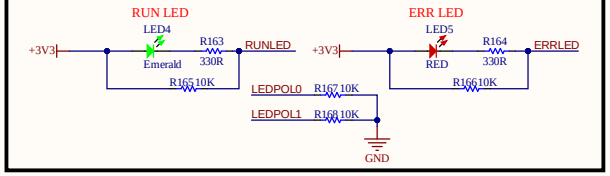




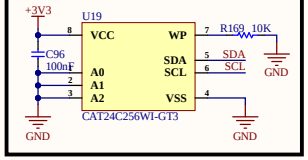
RJ45

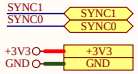


EARTH TERMINAL



EEPROM

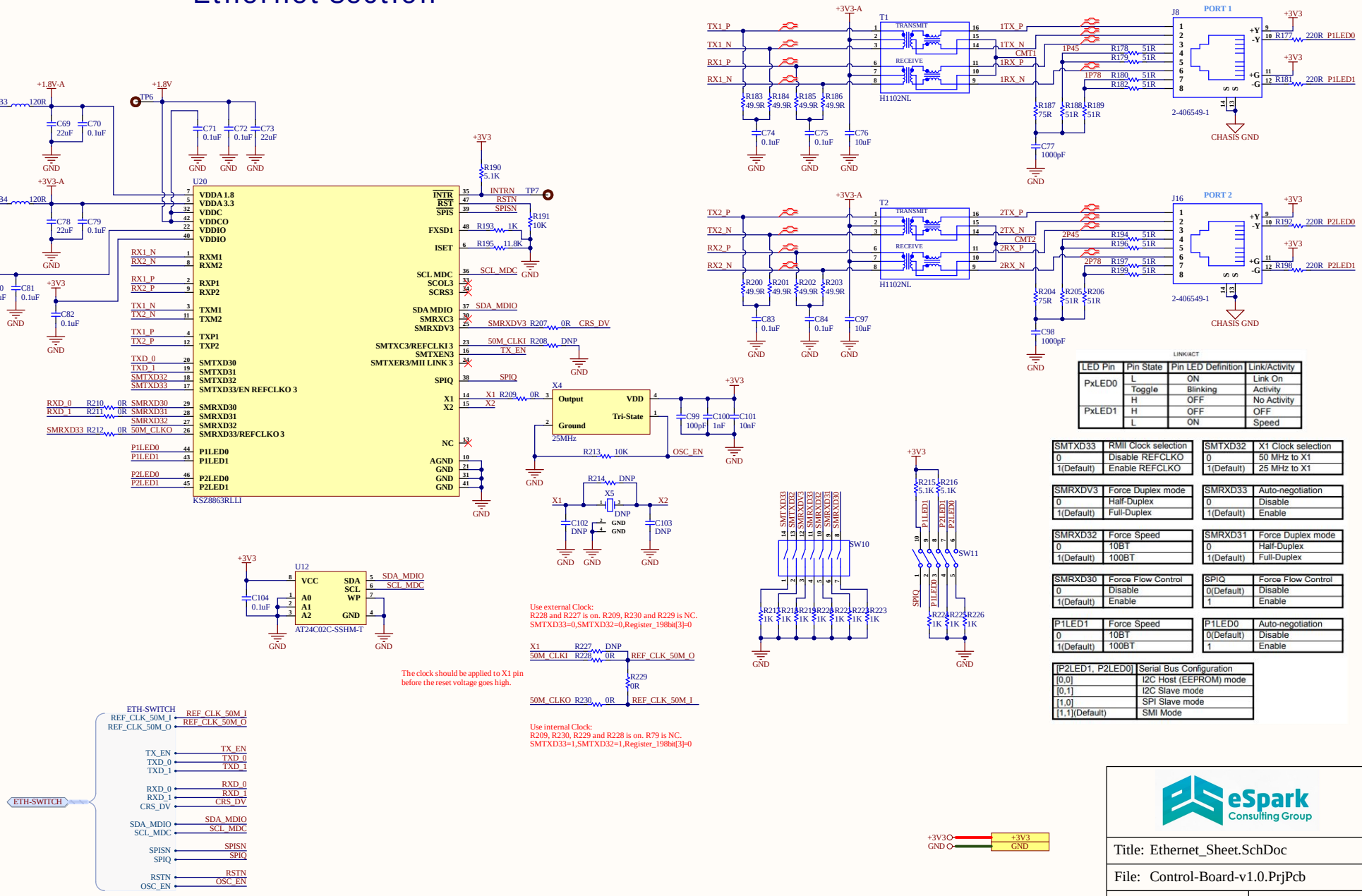






Title: EtherCAT_Sheet.SchDoc	
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Ethernet Section



LED Pin	Pin State	Pin LED Definition	Link/Activity
PxLED0	L	ON	Link On
	Toggle	Blinking	Activity
	H	OFF	No Activity
PxLED1	H	OFF	OFF
	L	ON	Speed

LINK/ACT

SMTXD33	RMII Clock selection
0	Disable REFCLKO
1(Default)	Enable REFCLKO

SMTXD32	X1 Clock selection
0	50 MHz to X1
1(Default)	25 MHz to X1

SMRXDV3	Force Duplex mode
0	Half-Duplex
1(Default)	Full-Duplex

SMRXD33	Auto-negotiation
0	Disable
1(Default)	Enable

SMRXD32	Force Speed
0	10BT
1(Default)	100BT

SMRXD31	Force Duplex mode
0	Half-Duplex
1(Default)	Full-Duplex

SMRXD30	Force Flow Control
0	Disable
1(Default)	Enable

SPIQ	Force Flow Control
0(Default)	Disable
1	Enable

P1LED1	Force Speed
0	10BT
1(Default)	100BT

P1LED0	Auto-negotiation
0(Default)	Disable
1	Enable

[P2LED1, P2LED0]	Serial Bus Configuration
[0,0]	I2C Host (EEPROM) mode
[0,1]	I2C Slave mode
[1,0]	SPI Slave mode
[1,1](Default)	SMI Mode

Use external Clock:
R228 and R227 is on. R209, R230 and R229 is NC.
SMTXD33=0,SMTXD32=0,Register_198bit[3]=0

Use internal Clock:
R209, R230, R229 and R228 is on. R79 is NC.
SMTXD33=1,SMTXD32=1,Register_198bit[3]=0

The clock should be applied to X1 pin before the reset voltage goes high.

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